

1 Background and Notations

1.1 Poisson Processes

Definition 1. A counting process $N = (N(t))_{t \geq 0}$ is a sequence of non-negative random variables such that :

- $N(0) = 0$,
- $N(t)$ takes values in the set of non-negative integers,
- $N(t)$ is non-decreasing and right-continuous.

Definition 2. A Poisson process with rate $\lambda > 0$ is a counting process $N = (N(t))_{t \geq 0}$ such that :

- $N(0) = 0$,
- The process has independent increments,
- The number of events in any interval of length t follows a Poisson distribution with parameter λt .

1.2 Cramér-Lundberg Model

We formally define the surplus process as follows :

$$R(t) = u + ct - S(t), \quad (1)$$

with u the initial reserve, c the premium rate, and $S(t)$ the aggregate claims up to time t .

In order to model $S(t)$, we consider the classical Cramér-Lundberg model which assumes that :

$$S(t) = \sum_{i=1}^{N(t)} X_i, \quad (2)$$

where $N(t)$ is a Poisson process with rate $\lambda > 0$, representing the number of claims up to time t . The X_i are i.i.d. random variables representing the claim sizes, **independent** of $N(t)$, with common distribution F_X .